

AOE 6444: Homework 1

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1 Monte Carlo Sampling

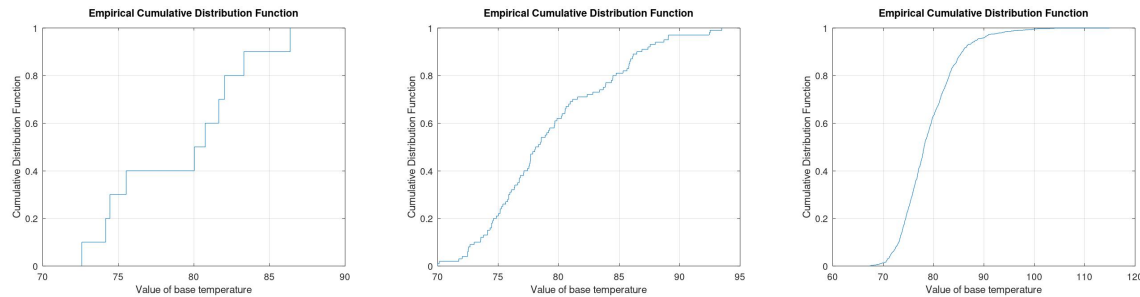


Figure 1: Cumulative Distributive Functions of base temperature using 10, 100, 1000 samples respectively (left to right).

Based on the plots from Figure 1, when 10 samples are used, the probability that the CPU base temperature will exceed the peak allowable temperature of 85 degrees is 10%. This function is more jagged due to the smaller sample size, meaning the base temperature varies a lot depending on which samples I choose. The reported plot is from the third trial run. The first 2 runs did not give me any base temperatures that exceeded 85 degrees. When the sample size increases to 100, the probability that the CPU base temperature will exceed the peak allowable temperature of 85 degrees is 19%. Compared to the sample size of 10, this is more smooth but still has some variability. Lastly, when sample size increases to 1000, you see a much more smoother function and stable. Here, the probability that the CPU base temperature will exceed the peak allowable temperature of 85 degrees is 12.2%.

2 Latin Hypercube Sampling

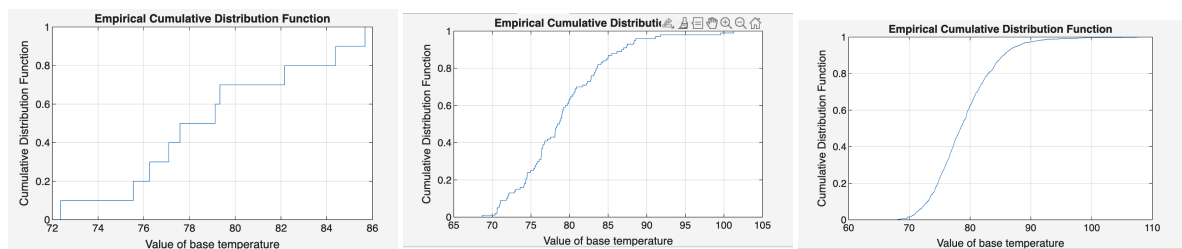


Figure 2: Cumulative Distributive Functions of base temperature using 10, 100, 1000 samples respectively (left to right).

Based on the plots from Figure 2, when 10 samples are used, the probability that the CPU base temperature will exceed the peak allowable temperature of 85 degrees is 10%. This is less jagged compared to Monte Carlo sampling and appears more uniform. When the sample size increases to 100, the probability that the base temperature will exceed the peak allowable temperature of 85 degrees is 14%. It demonstrates faster convergence than the respective Monte Carlo estimate. Lastly, when sample size increases to 1000, the probability that the base temperature will exceed the peak allowable temperature of 85 degrees is 11.4%.